



ODEX – Smart OD Management & Approval System
Project report submitted in Partial fulfilment of the requirement for the
Award of the degree of

BACHELOR OF COMPUTER APPLICATIONS (BCA)
BANGALORE UNIVERSITY

Submitted by

NAME: Syed Awaiz

Reg. No: U03EH23S0082

Under the guidance of

Dr. Sunetra Chatterjee

Assistant Professor

IFIM School of Technology



IFIM

IFIM COLLEGE (AUTONOMOUS)
8P & 9P, KIADB Industrial Area,
Electronics City Phase I, Bengaluru - 560100



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CERTIFICATE FROM THE GUIDE

This is to certify that the dissertation titled **“ODEX – Smart OD Management & Approval System”** is the Bonafide work which is carried out by **Syed Awaiz** bearing **Register no: U03EH23S0082** in partial fulfilment of the requirement for the award of BCA degree of Bangalore University, under my Guidance and Supervision.

The Project report submitted by her has been successfully completed and reflects her hard work and sincere effort.

Place: Bengaluru

Date:

Dr. Sunetra Chatterjee
Project Guide



CERTIFICATE OF ORIGINALITY

This is to certify that the project titled “**ODEX – Smart OD Management & Approval System**” is an original work of **Syed Awaiz** bearing **Register no: U03EH23S0082** and is submitted in the partial fulfilment of the requirement of the Bangalore University for the award of Bachelor of Computer Applications. The work was not submitted earlier to this University/Institution for the fulfilment of the requirement of a course of study. **Mr. Syed Awaiz** is guided by **Dr. Sunetra Chatterjee**, the faculty guide per the regulations of Bangalore University.

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Dr. Vishal C

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Dr. Salur Srikant Patnaik

Place: Bengaluru

Date:



CERTIFICATE FROM THE COMPANY

This is to certify that **Mr. Syed Awaiz**, bearing Register Number **U03EH23S0082**, a student of Bachelor of Computer Applications (BCA), Bangalore University, has successfully completed the project titled **“ODEX – Smart OD Management & Approval System”** during the academic year **2023–2026** as a partial fulfilment for the award of the degree of Bachelor of Computer Applications.

The project work was carried out at **IFIM College (Autonomous)** under the guidance of **Dr. Sunetra Chatterjee**. During the project, the student has demonstrated commendable dedication, technical competency, sincerity, and professionalism in the areas of workflow automation, frontend-backend integration, API development, and institutional management systems.

We hereby acknowledge that the project submitted by the student is a genuine and original piece of work completed to the satisfaction of the institution and in accordance with the academic requirements prescribed by Bangalore University.

Place: Bengaluru

Date:

Dr. Sunetra Chatterjee

Project Mentor

Dr. Salur Srikant Patnaik

Dean- IFIM School of Technology



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I am grateful to **Dr. Sunetra Chatterjee**, my external company guide, who gave their valuable time for the interaction and allowed me to carry out this project.

I would also like to thank all who helped me directly or indirectly in completing this project successfully.

Name: Syed Awaiz

Reg. No: U03EH23S0082



DECLARATION

I hereby declare that the dissertation titled “**ODEX – Smart OD Management & Approval System**” submitted in partial fulfilment of the requirement of the degree of Bachelor of Computer Applications in Bangalore University has been prepared by me during the academic year **2023-26** under the guidance of **Dr. Sunetra Chatterjee, Assistant Professor**, School of Technology, IFIM College.

I further declare that this Project Report is the outcome of my own efforts and that is not submitted to any other University or Institute for the award of other degree or diploma or other certificate.

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Register No: U03EH23S0082

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EXECUTIVE SUMMARY

The ODEX (Smart OD Management & Approval System) is a web-based workflow automation platform developed to simplify and digitize the traditional On-Duty (OD) approval process followed in educational institutions. The project was designed to overcome the limitations of manual paper-based approval systems such as delayed processing, excessive paperwork, lack of transparency, and manual attendance updates.

The system provides a centralized platform where students can submit OD requests digitally and track their approval status in real time. Faculty members, Heads of Departments (HODs), and administrators can review, approve, reject, and monitor requests through role-based dashboards. The platform also integrates attendance management to automatically synchronize approved OD requests with attendance records.

The frontend of the ODEX system was developed using React.js to provide a responsive and interactive dashboard-based user interface. The backend was developed using Python Flask framework with REST API architecture for request processing, workflow automation, authentication handling, and database communication. PostgreSQL was used as the centralized database management system for storing users, requests, approvals, attendance records, and workflow history.

The project implements several important features such as:

- Secure role-based authentication
- Online OD request submission
- Real-time workflow tracking
- Attendance synchronization
- Dashboard-based communication
- Approval workflow automation
- REST API integration
- Centralized database management

The backend workflow engine automates request routing between faculty members, HODs, and administrators while dynamically updating workflow status and notifications. The system

significantly reduces manual workload and improves communication efficiency between departments.

Different testing methodologies such as unit testing, integration testing, authentication testing, database testing, and system testing were performed successfully to ensure proper functionality, reliability, and security of the platform.

Overall, the ODEX platform successfully provides a scalable, secure, and efficient digital workflow management system capable of improving institutional transparency, workflow automation, attendance management, and approval processing through modern frontend-backend integration technologies.

CHAPTER 1: INTRODUCTION

1.1 Introduction

In today's digital era, educational institutions are increasingly adopting automated systems to improve workflow efficiency, communication, and data management. However, many colleges still rely on traditional paper-based systems for managing On-Duty (OD) requests. Students who participate in technical events, sports competitions, industrial visits, conferences, cultural activities, hackathons, or medical emergencies are required to submit OD applications manually to faculty members and Heads of Departments (HODs).

The manual OD approval process involves multiple stages such as:

- Filling physical forms
- Collecting faculty signatures
- Submitting forms to HODs
- Verifying attendance records
- Updating departmental records manually

This traditional system creates several operational challenges including:

- Delayed approvals
- Increased paperwork
- Poor workflow transparency
- Difficult request tracking
- Human errors in attendance management
- Increased administrative workload

To overcome these limitations, the project titled **“ODEX – Smart OD Management & Approval System”** was developed as a modern web-based workflow automation platform.

The ODEX platform automates the complete OD management process digitally through:

- Frontend dashboards
- Backend APIs

- Workflow automation
- Authentication systems
- Attendance integration
- Approval tracking systems

The project was designed using modern full-stack technologies to improve communication between students, faculty members, HODs, and administrators.

The frontend system was developed using **React.js** to provide interactive user dashboards and real-time workflow monitoring.

The backend system was developed using **Python Flask Framework** with REST API architecture to support:

- Authentication
- Workflow processing
- Request validation
- Role-based authorization
- Approval management
- Database communication

The system significantly improves workflow efficiency and reduces dependency on manual paperwork.

1.2 Background of the Project

Educational institutions conduct several academic and non-academic activities where students require attendance exemptions or On-Duty permissions. Traditionally, OD requests are managed through paper forms and physical approval systems.

The manual approach creates several difficulties:

- Students must move physically between departments
- Faculty members manually verify requests
- HOD approvals consume additional time

- Attendance updates are delayed
- Tracking request status becomes difficult

As institutions grow larger, the number of OD requests increases significantly, making manual workflow management inefficient.

To solve these issues, institutions require a centralized automated system capable of:

- Handling OD requests digitally
- Automating workflow approvals
- Providing real-time request tracking
- Integrating attendance systems
- Reducing paperwork

The ODEX system was developed as a solution to these institutional workflow challenges.

1.3 Need for the Project

The traditional OD management process lacks automation and centralized digital workflow management.

The major problems identified in the existing system include:

1. Delayed Approval Process

Students must manually collect approvals from multiple authorities which consumes more time.

2. Increased Paperwork

Physical forms increase administrative burden and document management difficulties.

3. Lack of Transparency

Students cannot track request progress dynamically.

4. Attendance Management Issues

Attendance updates are delayed because records are maintained manually.

5. Poor Communication

Communication between departments becomes slow and inefficient.

6. Difficult Record Maintenance

Managing historical OD records manually creates storage and retrieval challenges.

7. Lack of Real-Time Tracking

The manual system does not provide live workflow status updates.

To overcome these challenges, there was a need to develop a centralized digital workflow system capable of automating OD request management.

1.4 Objectives of the Project

The major objectives of the ODEX system are:

Primary Objectives

- To automate the traditional OD approval workflow
- To develop a centralized web-based OD management platform
- To reduce manual paperwork and administrative workload
- To improve request tracking transparency
- To provide secure authentication mechanisms
- To integrate attendance management with OD workflows

Technical Objectives

- To implement REST API architecture
- To develop frontend-backend integration
- To implement role-based access control
- To improve workflow synchronization
- To provide scalable backend architecture

User-Oriented Objectives

- To simplify OD request submission
- To provide real-time status monitoring

- To improve communication between users
- To provide dashboard-based workflow visualization

1.5 Scope of the Project

The scope of the ODEX system includes the development of a complete digital workflow management platform for educational institutions.

The project mainly focuses on:

- OD request automation
- Attendance integration
- Dashboard management
- Authentication systems
- Workflow processing
- API communication
- Role-based authorization

The system supports multiple users including:

- Students
- Faculty members
- Heads of Departments
- Administrators

The backend system manages:

- Request validation
- Approval routing
- API response handling
- Workflow synchronization
- Database communication

The frontend system provides:

- Dashboard interfaces
- Workflow visualization
- Attendance analytics
- Real-time status tracking

1.6 Problem Statement

Many educational institutions continue to depend on manual OD approval systems that are inefficient, time-consuming, and difficult to manage.

The existing manual workflow creates:

- Approval delays
- Poor transparency
- Increased administrative burden
- Attendance update problems
- Difficult workflow monitoring

Therefore, there is a need to develop a modern web-based platform capable of automating the OD workflow digitally while improving transparency, communication, and operational efficiency.

The ODEX system was developed to solve these institutional workflow challenges through centralized backend-driven automation.

1.7 Proposed Solution

The proposed solution is the development of a centralized web-based OD management system capable of automating approval workflows digitally.

The proposed system includes:

- React frontend dashboards
- Flask backend APIs
- Authentication systems
- Workflow automation modules

- Attendance integration
- Real-time request tracking

The system processes workflow operations dynamically and provides dashboard-based communication between users.

The proposed architecture improves:

- Workflow transparency
- Approval efficiency
- Communication speed
- Attendance synchronization
- Request management

1.8 Advantages of the Proposed System

The ODEX platform provides several advantages over traditional manual systems.

Major Advantages

- Automated workflow processing
- Faster approval management
- Centralized request handling
- Improved workflow transparency
- Reduced paperwork
- Better attendance management
- Real-time status updates
- Improved security and validation
- Efficient frontend-backend communication
- Better scalability

Institutional Benefits

- Reduced administrative workload

- Better workflow coordination
- Improved operational efficiency
- Better student experience

1.9 Modules of the System

The ODEX platform consists of multiple integrated modules responsible for workflow automation.

1. Login Module

Handles:

- Authentication
- Password verification
- User role identification

2. Student Dashboard Module

Provides:

- OD request submission
- Request tracking
- Attendance monitoring

3. Faculty Module

Allows faculty members to:

- Review OD requests
- Approve or reject requests

4. HOD Module

Handles:

- Department approval
- Final workflow verification

5. Attendance Module

Provides:

- Attendance analytics
- Subject-wise attendance tracking
- OD-adjusted attendance updates

6. Backend API Module

Handles:

- Request validation
- Workflow processing
- Database communication
- Response generation

1.10 Technology Stack Used

The ODEX system was developed using modern technologies capable of supporting scalable workflow automation.

Table 1.1: Software Technologies Used in ODEX System

Technology	Purpose
React.js	Frontend Development
Python	Backend Programming
Flask	Backend Framework
REST APIs	Communication Layer
JSON	Data Exchange
VS Code	Development Environment
Postman	API Testing
GitHub	Version Control
Chrome Browser	Frontend Testing

1.11 Frontend Contribution Overview

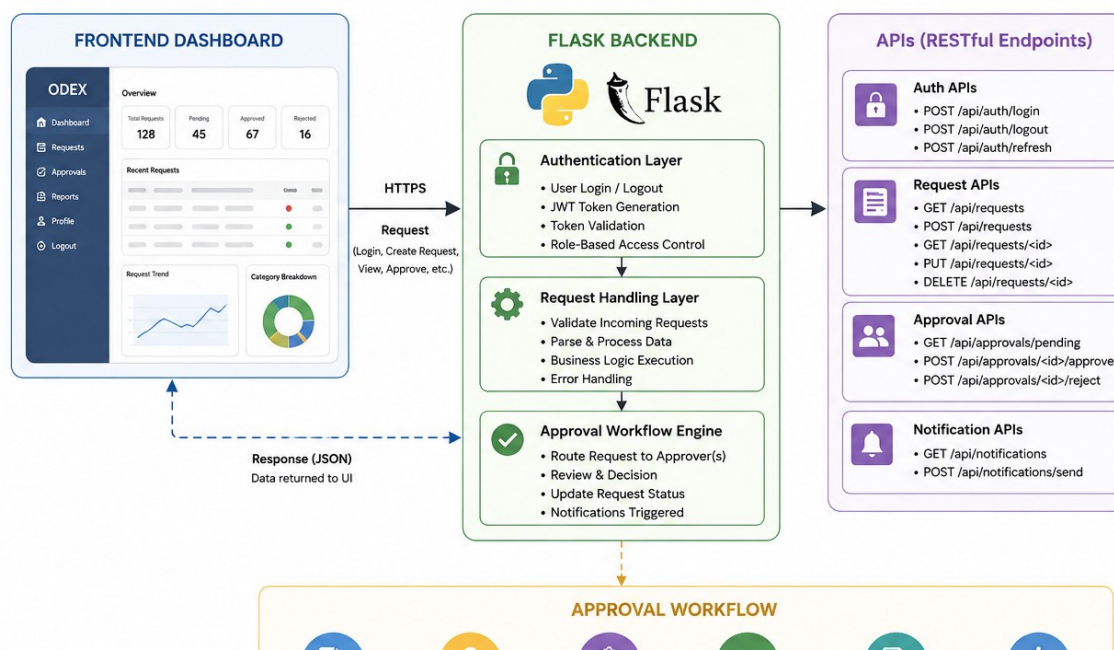
The frontend system provides interactive dashboards and workflow visualization.

The frontend contribution includes:

- Login page design
- Dashboard development
- Attendance monitoring interface
- Workflow tracking system
- Request submission forms
- Frontend-backend API integration
- Responsive UI development

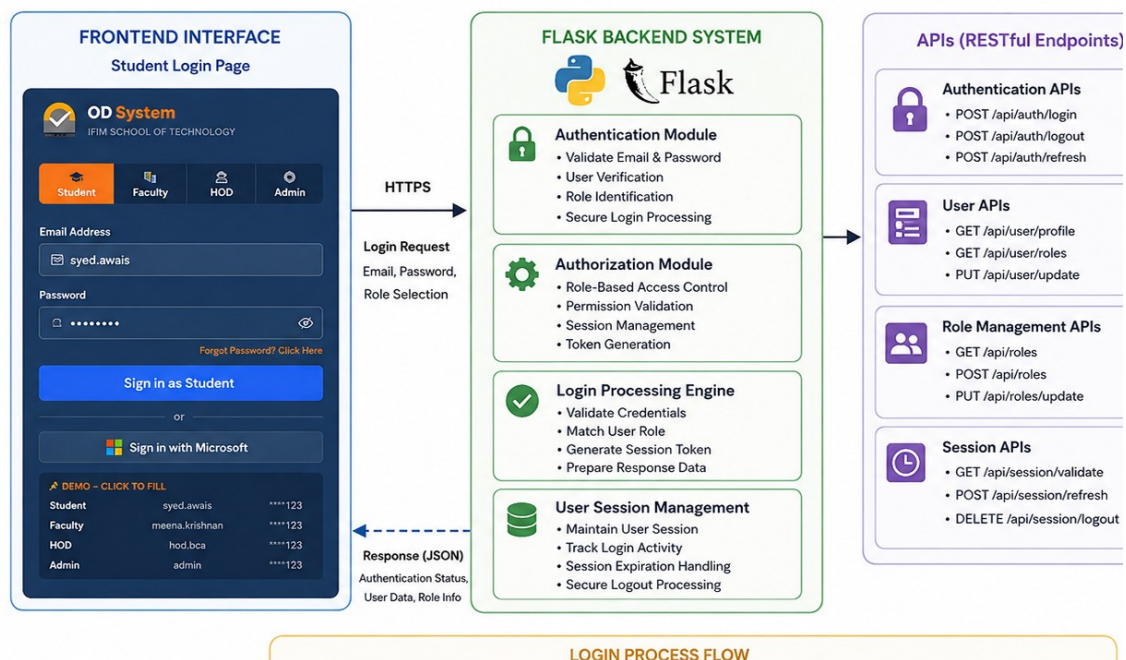
The frontend communicates with backend APIs using REST architecture and JSON data exchange.

Figure 1.1: ODEX Backend Architecture Overview



- The ODEX backend architecture connects the frontend dashboard, Flask backend, APIs, and database for centralized workflow management.
- The authentication layer validates users and provides secure role-based access control for students, faculty, HODs, and admins.
- The request handling layer processes OD requests, validates data, and manages workflow operations efficiently.
- REST APIs enable smooth communication between frontend dashboards and backend services using JSON responses.
- The PostgreSQL database stores user records, OD requests, approvals, notifications, and workflow history securely.

Figure 1.2: ODEX Student Login Interface Overview



- The student login interface provides secure authentication access to the ODEX platform using email and password validation.
- The system supports multiple user roles including Student, Faculty, HOD, and Admin through role-based login selection.

- The login page includes features such as Microsoft sign-in integration, forgot password option, and demo credentials for testing.
- The frontend interface communicates with the Flask backend APIs to validate user credentials and authorize dashboard access.
- The login module improves security, centralized access management, and smooth navigation to the respective user dashboards.

Chapter 2: SYSTEM ANALYSIS

2.1 Introduction

System analysis is one of the most important phases in software development. It involves studying the existing system, identifying operational limitations, understanding workflow requirements, and designing an improved solution capable of overcoming existing problems.

The system analysis phase helps developers understand:

- How data flows through the application
- How workflow operations are processed
- How backend systems communicate with frontend dashboards
- How automation can improve institutional workflow efficiency

The ODEX (Smart OD Management & Approval System) was developed to automate the traditional manual OD approval workflow using modern frontend and backend technologies.

The system analysis phase mainly focused on:

- Identifying limitations in the existing manual system
- Designing a centralized workflow architecture
- Improving approval processing efficiency
- Reducing paperwork
- Improving request tracking transparency

The proposed ODEX system provides:

- Dashboard-based communication
- REST API architecture
- Authentication systems
- Attendance integration
- Workflow automation

The analysis phase provided the foundation for implementing a scalable and efficient digital OD management platform.

2.2 Existing System

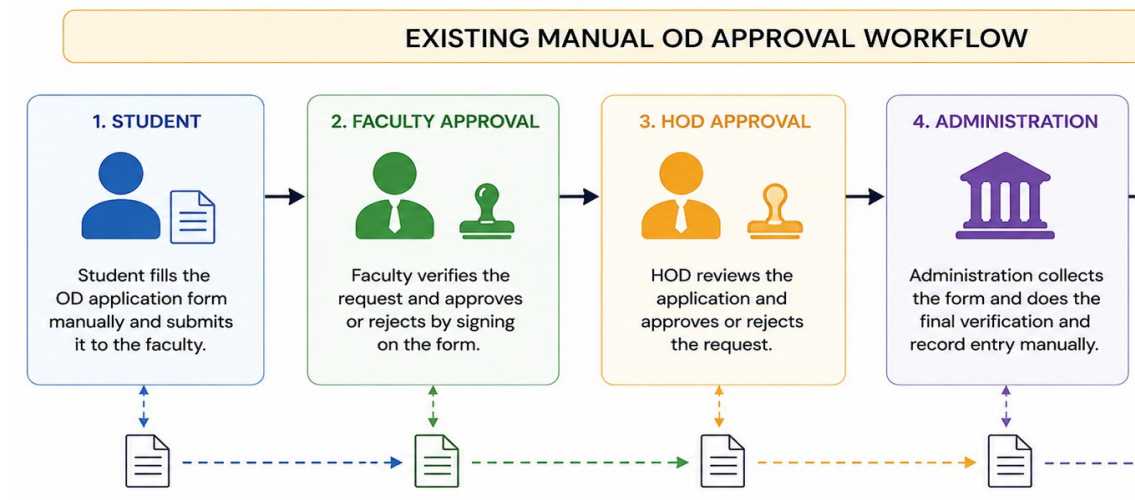
In many educational institutions, OD approval workflows are still handled manually through physical forms and paper-based documentation.

Students are required to:

1. Fill physical OD application forms
2. Collect faculty signatures
3. Submit forms to HODs
4. Verify attendance manually
5. Wait for administrative approval

The traditional workflow creates several operational difficulties and consumes significant time.

2.2.1 Workflow of Existing System



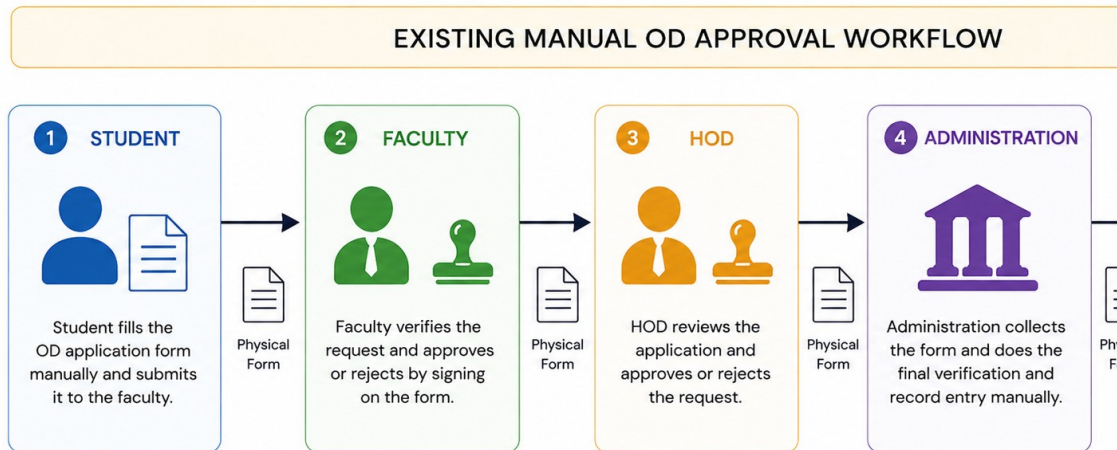
The traditional workflow depends on physical paper movement between departments.

Existing Workflow

- Student submits OD form
- Faculty verifies request

- HOD approves request
- Administration updates records
- Attendance is updated manually

Figure 2.1: Existing Manual Workflow System



2.3 Proposed System

The proposed ODEX platform automates the complete OD approval workflow digitally using frontend dashboards and Flask backend APIs.

The proposed system provides:

- Online OD request submission
- Dashboard-based tracking
- Authentication systems
- Automated approval workflows
- Attendance integration
- Real-time status updates

The system improves communication between students, faculty members, HODs, and administrators.

2.4 Feasibility Study

The ODEX system was analysed using different feasibility methods.

Technical Feasibility

Modern technologies such as React.js, Flask, and REST APIs support workflow automation and frontend-backend integration efficiently.

Economic Feasibility

The system uses open-source technologies, reducing development and maintenance cost.

Operational Feasibility

The platform successfully supports digital workflow automation and centralized request management.

2.5 Comparison Between Existing and Proposed System

Existing System	Proposed ODEX System
Manual workflow	Automated workflow
Paper-based forms	Digital request submission
Delayed approvals	Faster processing
Manual attendance updates	Automated attendance integration
Difficult tracking	Real-time tracking
Increased paperwork	Reduced paperwork

2.6 Summary

The system analysis phase helped identify the limitations of the traditional OD approval process and provided the foundation for developing the ODEX platform.

The proposed system successfully improves:

- Workflow automation
- Request processing

- Attendance synchronization
- Approval transparency
- Frontend-backend communication

The ODEX platform provides an efficient and centralized digital workflow management system for educational institutions.

CHAPTER 3: SYSTEM SPECIFICATION

3.1 Introduction

System specification defines the hardware and software requirements necessary for developing and executing the ODEX (Smart OD Management & Approval System). Proper system specification ensures smooth frontend rendering, backend processing, workflow automation, authentication handling, API communication, and database operations.

The ODEX system was developed using modern frontend and backend technologies capable of supporting:

- Dashboard-based communication
- REST API integration
- Attendance management
- Approval workflow automation
- Frontend-backend synchronization

This chapter explains the hardware and software requirements used during the development and testing of the ODEX platform.

3.2 Hardware Used

The ODEX platform was developed using a computer system capable of supporting frontend development, backend execution, browser testing, API communication, and workflow processing.

The selected hardware configuration provided sufficient performance for:

- React frontend rendering
- Flask backend execution
- API testing
- Attendance dashboard processing
- Workflow synchronization

3.2.1 Hardware Requirements

Component	Specification
Processor	Intel Core i5 or Higher
Storage	256 GB SSD
Monitor	15-inch Display
Keyboard	Standard Keyboard
Mouse	Optical Mouse
Internet Connection	Broadband Internet

3.2.2 Processor

The project was developed using an Intel Core i5 processor capable of supporting backend execution, frontend rendering, and multitasking operations.

The processor handled:

- Flask server execution
- React development environment
- Browser testing
- API communication
- Workflow processing

A multi-core processor improved overall system performance during development and testing.

3.2.3 Memory (RAM)

The system used 8 GB RAM to support:

- React frontend rendering
- Backend processing
- VS Code development environment

- Postman API testing
- Browser execution
- Workflow synchronization

Adequate RAM improved multitasking and frontend-backend integration performance.

3.2.4 Storage

SSD storage was used for:

- Faster project loading
- Quick dependency installation
- Efficient backend execution
- Better file management

The project contains:

- Frontend components
- Backend APIs
- Route modules
- Attendance dashboards
- Configuration files

SSD storage improved development efficiency significantly.

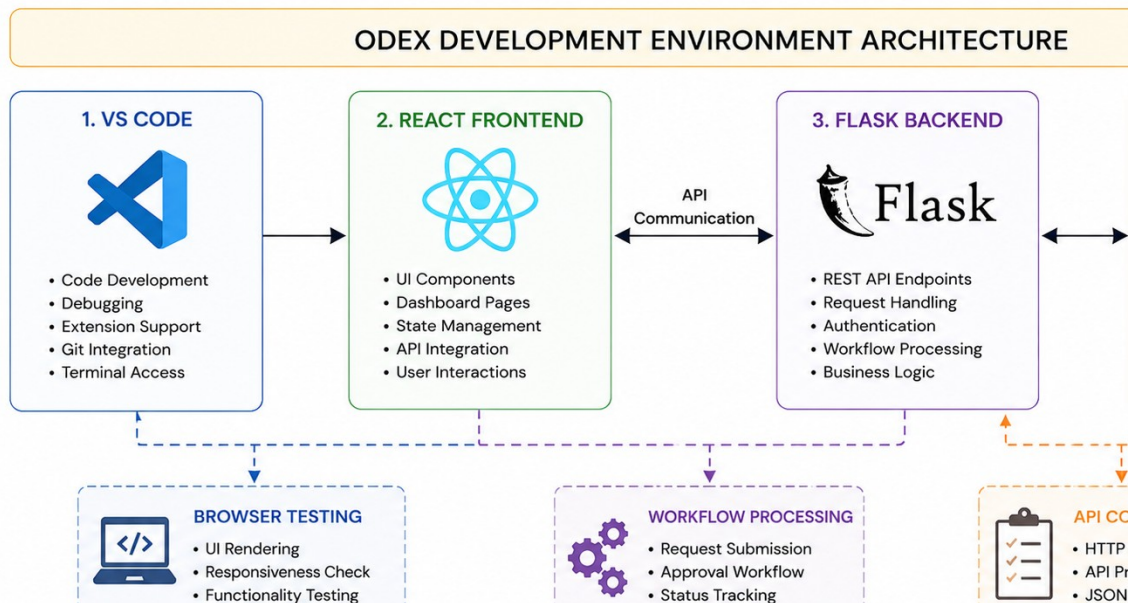
3.2.5 Internet Connectivity

Internet connectivity was required for:

- Installing React packages
- Installing Python libraries
- Accessing API documentation
- GitHub version control
- Framework updates

Broadband internet improved package installation and project synchronization.

Figure 3.1: Development Environment Setup



Include:

- API Communication
- Browser Testing
- Workflow Processing

3.3 Software Used

The ODEX system was developed using modern software technologies capable of supporting workflow automation and frontend-backend integration.

The selected technologies enabled:

- Authentication systems
- Dashboard rendering
- API communication
- Attendance management
- Workflow synchronization

3.3.1 Software Requirements

Software	Purpose
Windows 10/11	Operating System
React.js	Frontend Development
Python	Backend Programming
Flask	Backend Framework
VS Code	Code Editor
Postman	API Testing
JSON	Data Exchange
GitHub	Version Control
Chrome Browser	Frontend Testing
Node.js	React Runtime Environment

3.3.2 Operating System

The ODEX platform was developed using Windows operating system.

The operating system supported:

- React.js frontend environment
- Python backend execution
- Flask framework
- Browser testing
- API debugging

Windows provided smooth execution of frontend and backend development tools.

3.3.3 Visual Studio Code

Visual Studio Code (VS Code) was used as the primary code editor for frontend and backend development.

Features Used

- Syntax highlighting
- Integrated terminal
- Extension support
- Git integration
- File management
- Python debugging
- React development support

VS Code improved coding efficiency and project organization.

3.3.4 React.js

React.js was used for frontend development of the ODEX platform.

React.js provided:

- Dynamic UI rendering
- Reusable components
- Faster frontend updates
- Dashboard-based architecture
- State management support

The frontend includes:

- Login interface
- Student dashboard
- Attendance dashboard
- Workflow tracking system

- Request submission module

3.3.5 Python Programming Language

Python was used as the primary backend programming language.

Python was selected because of:

- Simplicity
- Scalability
- Readability
- Flask compatibility
- Strong backend support

Python enabled implementation of:

- Workflow automation
- Authentication systems
- Backend APIs
- Request validation
- Attendance synchronization

3.3.6 Flask Framework

Flask is a lightweight Python web framework used for backend API development.

Flask was selected because of:

- Lightweight architecture
- Flexible routing system
- REST API support
- Easy frontend integration
- Efficient request handling

Flask was used to implement:

- Login APIs
- Approval workflows
- Attendance APIs
- Request validation systems
- Role-based authentication

3.3.7 JSON Data Format

JSON (JavaScript Object Notation) was used as the primary data exchange format between frontend dashboards and backend APIs.

JSON was used for:

- API requests
- API responses
- Attendance updates
- Login data handling
- Workflow communication

JSON improved frontend-backend communication efficiency.

3.3.8 Postman API Testing Tool

Postman was used for testing backend APIs and verifying server responses.

API Testing Included

- Login API testing
- Request submission testing
- Approval workflow testing
- Authentication validation
- JSON response verification

Postman improved backend debugging and workflow testing.

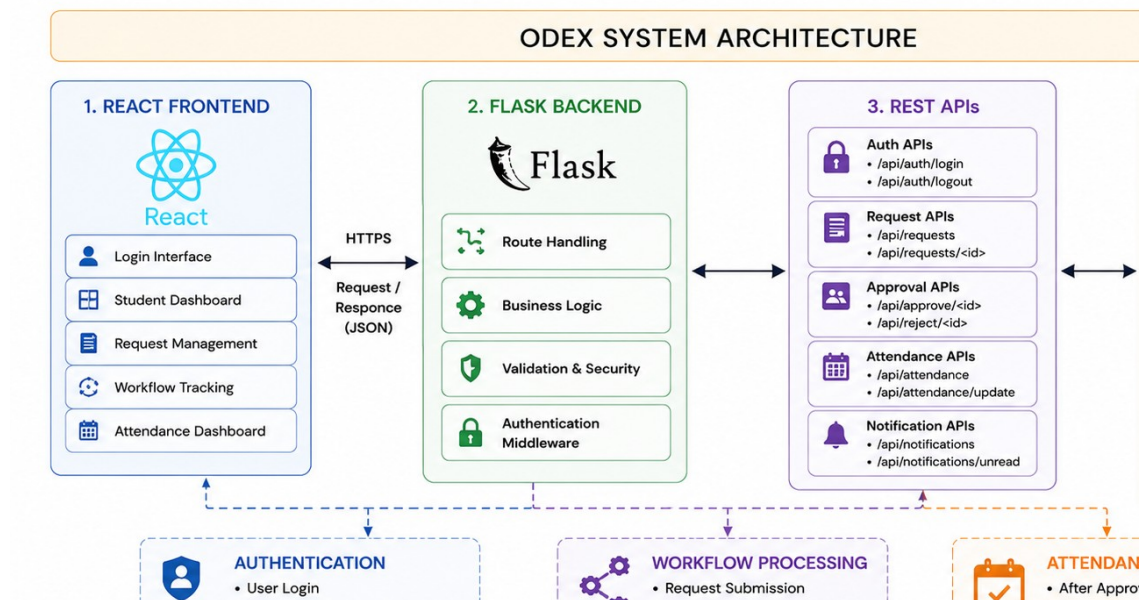
3.3.9 GitHub Version Control

GitHub was used for:

- Source code management
- Version tracking
- Project backup
- Collaboration support

GitHub improved project organization and backend maintenance.

Figure 3.2: Frontend-Backend Architecture



Include:

- Authentication
- Workflow Processing
- Attendance Updates

3.4 Development Environment

The ODEX development environment consisted of:

- React frontend environment

- Flask backend server
- REST API architecture
- JSON communication system
- VS Code editor
- Postman testing environment

The modular architecture improved scalability and maintainability.

3.5 Advantages of Selected Technologies

The selected technologies provided several advantages:

- Faster development process
- Better scalability
- Efficient API communication
- Improved workflow automation
- Better frontend rendering
- Simplified backend processing
- Better maintainability
- Secure authentication handling

These technologies were highly suitable for developing the ODEX workflow automation platform.

3.6 Summary

This chapter explained the hardware and software specifications used during the development of the ODEX platform.

Modern technologies such as React.js, Python, Flask, REST APIs, and JSON enabled the implementation of a scalable workflow management system capable of supporting:

- Authentication
- Attendance management

CHAPTER 4: SOFTWARE DESCRIPTION

4.1 Introduction

Software description explains the frontend and backend technologies used for developing the ODEX (Smart OD Management & Approval System). The project was designed using modern full-stack technologies capable of supporting workflow automation, dashboard communication, attendance management, and API integration.

The ODEX platform was developed using:

- React.js for frontend development
- Python Flask for backend development
- REST APIs for communication
- JSON for data exchange
- PostgreSQL for database management

The selected technologies helped implement a scalable and efficient workflow management system for educational institutions.

4.2 Frontend Description

The frontend of the ODEX system provides an interactive dashboard-based interface for students, faculty members, HODs, and administrators.

The frontend was developed using React.js and supports:

- Login authentication
- Dashboard navigation
- OD request submission
- Workflow tracking
- Attendance monitoring
- Status updates

The frontend communicates with backend APIs using REST architecture and JSON communication.

4.2.1 Features of Frontend

The frontend provides several important features:

- Responsive user interface
- Role-based dashboards
- Real-time workflow tracking
- Attendance analytics
- Dynamic request updates
- Interactive navigation system
- API integration support

The frontend was designed to improve user experience and workflow visibility.

4.2.2 Student Login Interface

The student login interface provides secure authentication access to the ODEX platform.

The login page includes:

- Email validation
- Password authentication
- Role selection
- Forgot password support
- Microsoft sign-in integration

The login module communicates with backend APIs for user validation and dashboard authorization.

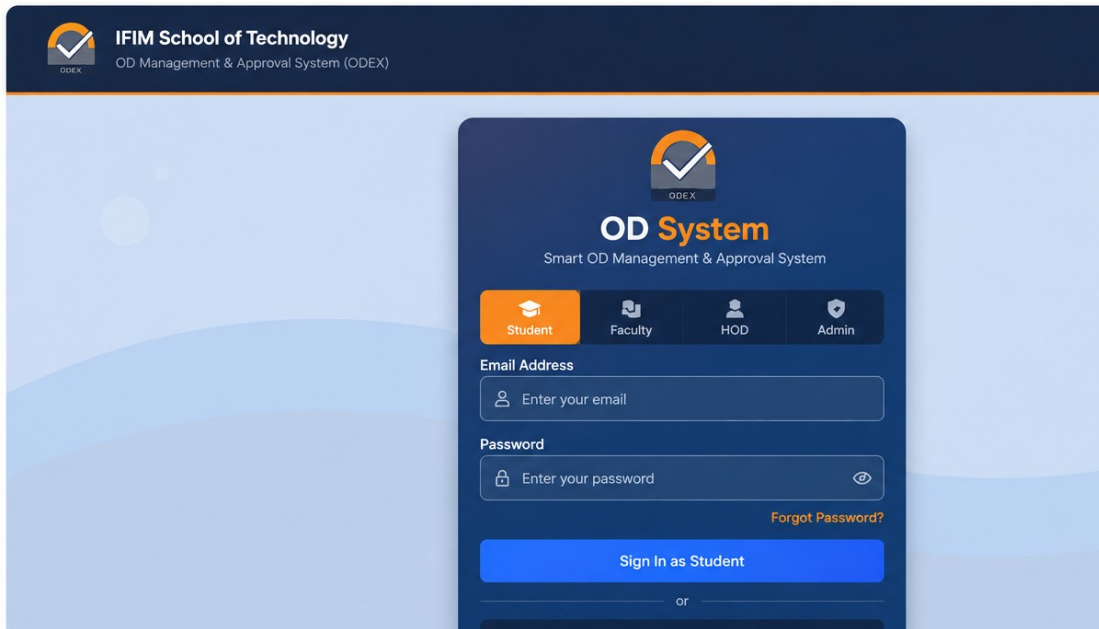
Figure 4.1: Student Login Interface

Description

The figure represents the secure login interface developed for students, faculty members, HODs, and administrators within the ODEX system.

Figure 4.1: Student Login Interface

The figure represents the secure login interface developed for students, faculty members, HODs, and administrators



4.2.3 Student Dashboard

The student dashboard acts as the central monitoring interface for OD requests and attendance management.

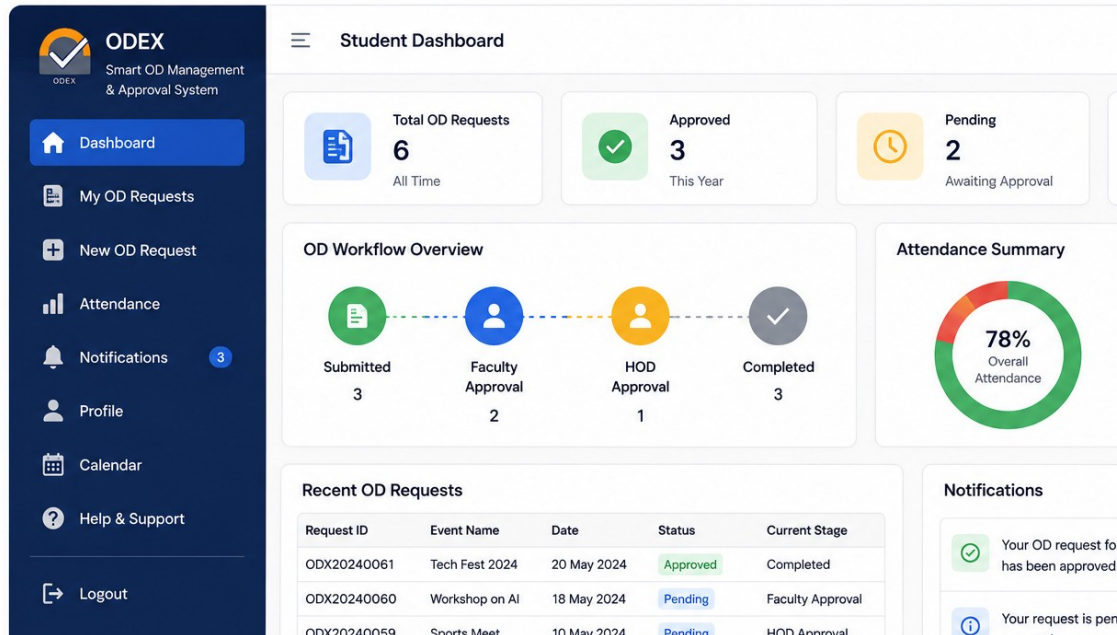
The dashboard displays:

- Active OD requests
- Approval workflow tracking
- Attendance summary
- Request statistics
- Workflow status updates

The dashboard dynamically receives information from backend APIs and updates workflow progress in real time.

Figure 4.2: Student Dashboard Interface

The dashboard provides centralized access to OD workflow tracking, attendance monitoring, and request management.



4.2.4 OD Request Submission Module

The OD request module allows students to submit OD applications digitally through a structured form interface.

The form captures:

- Student details
- Registration number
- Event type
- Event duration
- Venue information
- Organizer details

The submitted data is transmitted to backend APIs using JSON requests.

Figure 4.3: OD Request Submission Interface

The figure shows the digital OD request form used for workflow automation and online request submission.

ODEX Smart OD Management & Approval System

New OD Request

OD Request Submission Form
Fill in the details below to submit your OD request

Student Details

Student Name	Registration Number	Department	Semester
Arjun Sharma	IFIM2022CS123	Computer Science Engineering	6th Sem

Event Details

Event Title / Purpose	Event Type	Event Level
Tech Fest 2024	Technical	College Level

Event Start Date	Event End Date	Total OD Days	Venue
20/05/2024	21/05/2024	2	IFIM College Cam

Organizer Details

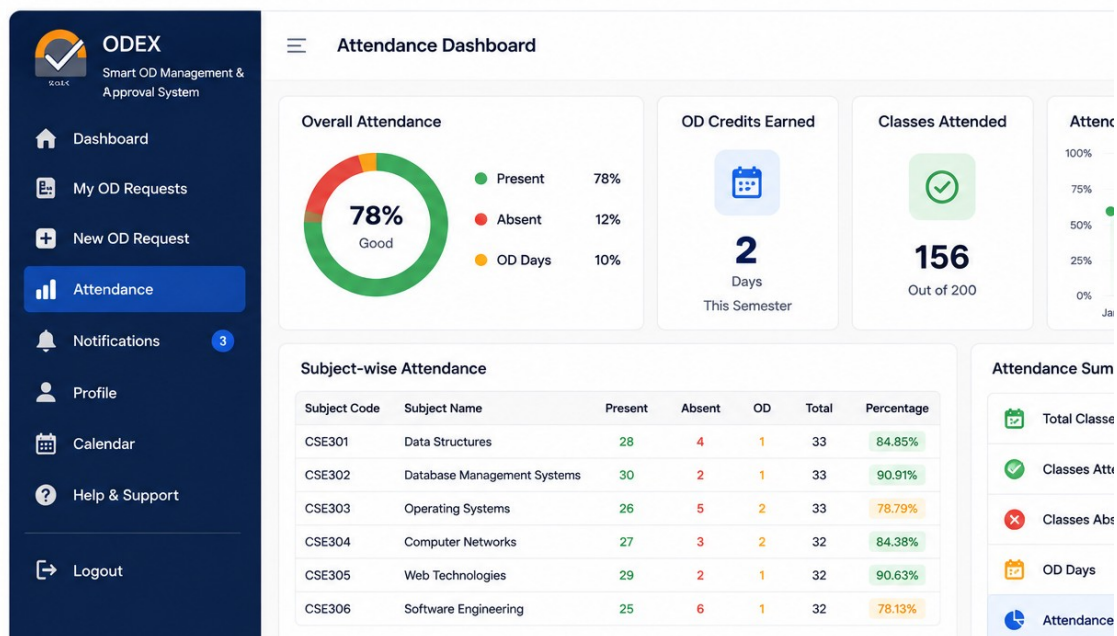
Organizer / Company	Organizer Contact	Organizer Email
IFIM College	080-12345678	techfest@ifim.edu

Additional Information

Description / Remarks	Upload Supporting Document (if any)
Participating in the technical event as a volunteer and presenter.	

Figure 4.5: Attendance Dashboard Interface

The attendance dashboard displays attendance statistics, OD credits, and subject-wise attendance.



4.3 Workflow Processing Engine

The workflow processing engine automates OD request routing between faculty members, HODs, and administrators.

The workflow sequence includes:

1. Student submits request
2. Faculty reviews request
3. HOD approval processing
4. Attendance update
5. Final status response

The backend dynamically updates workflow status and synchronizes dashboard information.

4.3.6 Database Communication

The backend communicates with the PostgreSQL database to store and retrieve workflow information.

The database stores:

- User details
- Roles
- OD requests
- Attendance records
- Approval history
- Notifications

Database integration ensures centralized workflow management and efficient record storage.

4.4 Frontend–Backend Integration

Frontend-backend integration enables smooth communication between dashboards and backend services.

The integration process includes:

- API request generation
- Backend request processing
- Database communication

- JSON response generation
- Frontend dashboard updates

The architecture supports:

- Authentication systems
- Workflow synchronization
- Attendance updates
- Real-time tracking

4.5 Summary

This chapter explained the frontend and backend software technologies used for developing the ODEX platform.

Modern technologies such as:

- React.js
- Flask Framework
- REST APIs
- JSON communication
- PostgreSQL database

enabled the implementation of a scalable and efficient OD workflow automation system.

The frontend dashboards and backend APIs successfully support:

- Authentication
- Attendance monitoring
- Workflow processing
- Approval management
- Real-time communication

The software architecture significantly improves workflow transparency and institutional workflow efficiency.

CHAPTER 5: PROJECT DESCRIPTION

5.1 Introduction

Project description explains the overall implementation, workflow architecture, modules, frontend-backend integration, database communication, and processing flow of the ODEX (Smart OD Management & Approval System).

The ODEX platform was developed as a centralized web-based workflow automation system for educational institutions. The system automates the complete On-Duty (OD) approval process digitally using modern frontend and backend technologies.

The project includes:

- Frontend dashboards
- Flask backend APIs
- Authentication systems
- Workflow automation
- Attendance management
- Approval tracking
- Database synchronization

The ODEX platform improves communication between students, faculty members, HODs, and administrators while reducing paperwork and workflow delays.

5.2 Project Overview

ODEX is a digital workflow management platform developed to automate institutional OD approval systems.

The system provides:

- Online OD request submission
- Real-time workflow tracking
- Attendance monitoring
- Dashboard-based communication

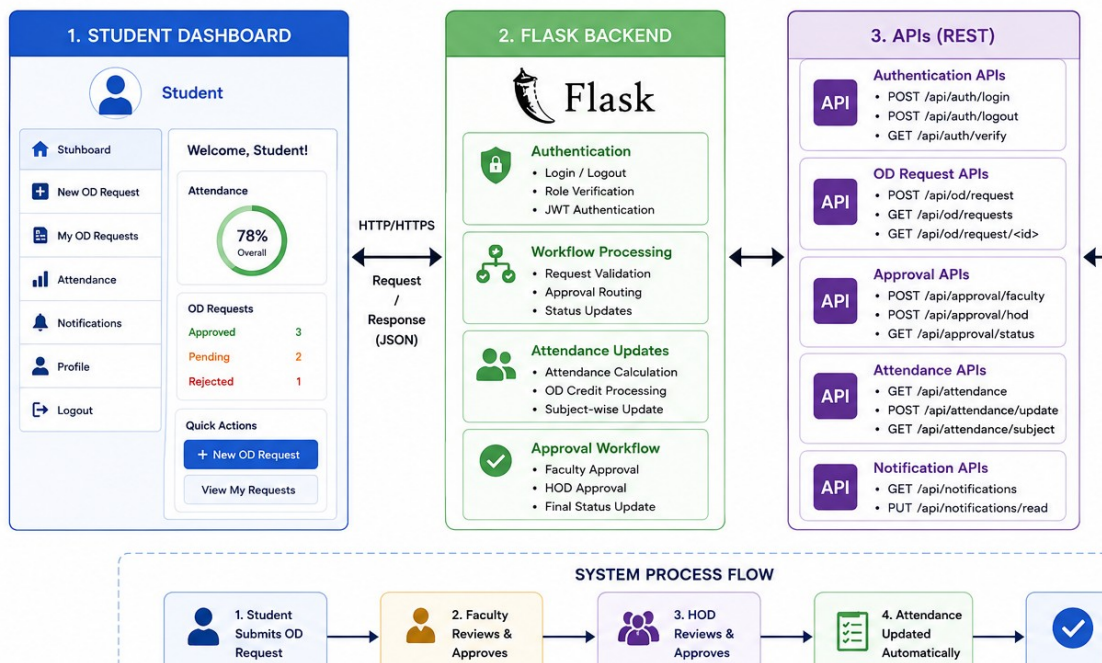
- Role-based authentication
- Approval workflow automation

The project architecture contains:

- React frontend dashboards
- Flask backend APIs
- PostgreSQL database
- REST API communication
- JSON data exchange

The system successfully integrates frontend and backend technologies to support centralized workflow management.

Figure 5.1: ODEX System Overview



Include:

- Authentication
- Workflow processing
- Attendance updates

- Approval workflow

5.3 Objectives of the Project

The major objectives of the ODEX platform are:

- To automate the manual OD approval process
- To reduce paperwork and administrative workload
- To provide real-time workflow tracking
- To implement centralized dashboard management
- To improve attendance synchronization
- To provide secure authentication systems
- To improve communication between departments
- To implement frontend-backend integration using REST APIs

The system was designed to improve institutional workflow efficiency and transparency.

5.4 System Modules

The ODEX platform consists of multiple integrated modules responsible for workflow automation and request management.

5.4.1 Login and Authentication Module

The login module provides secure access to the ODEX platform for:

- Students
- Faculty members
- HODs
- Administrators

The authentication system validates:

- Email credentials
- Passwords
- User roles

- Session authorization

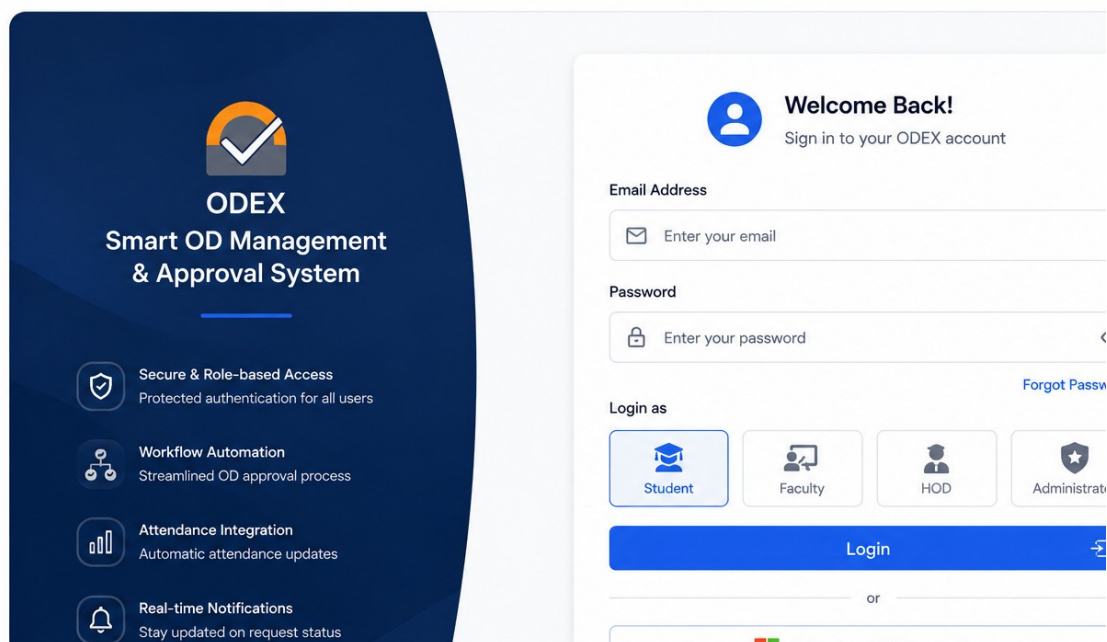
The module ensures secure dashboard access using backend authentication APIs.

Features

- Secure login validation
- Role-based access control
- Password authentication
- Session management
- Dashboard authorization

Figure 5.2: Student Login Interface

The figure represents the secure login interface developed for role-based authentication within the OD



5.4.2 Student Dashboard Module

The student dashboard acts as the main interface for workflow monitoring and attendance tracking.

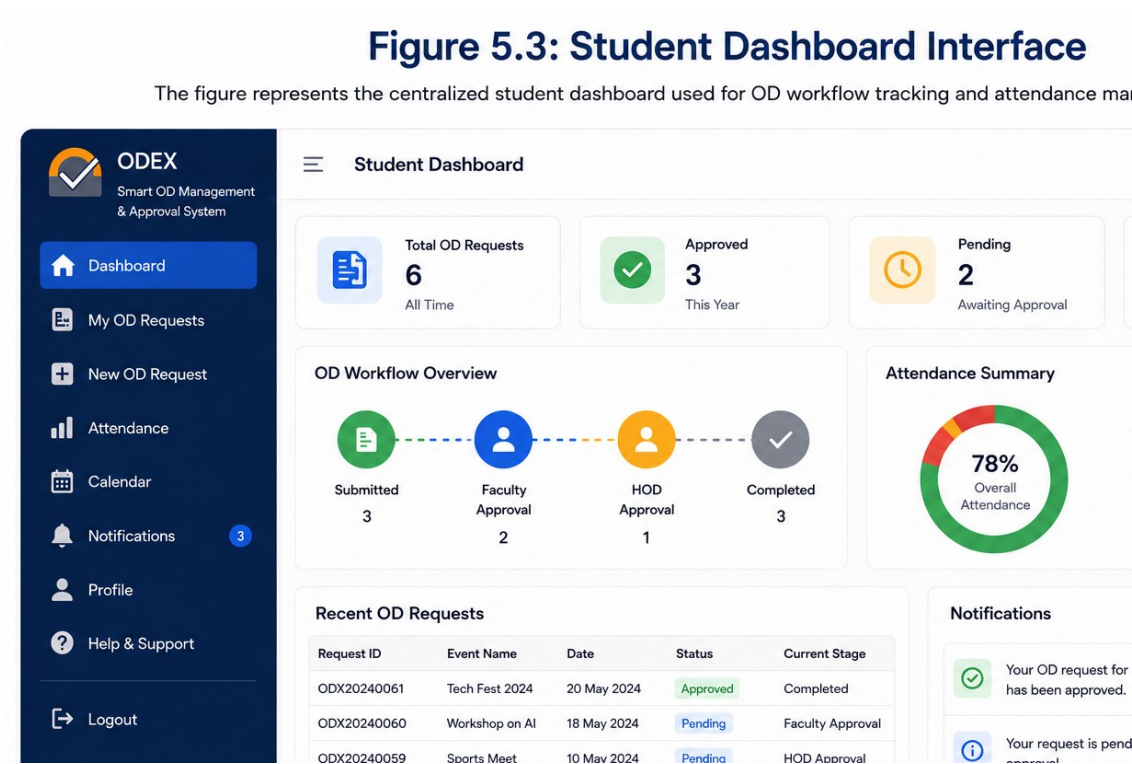
The dashboard provides:

- Active OD request tracking
- Attendance summary
- Request statistics
- Approval workflow monitoring
- Dashboard navigation

The frontend dashboard dynamically receives information from backend APIs using JSON responses.

Features

- Real-time status updates
- Attendance analytics
- Workflow visualization
- Dashboard communication



5.4.3 OD Request Submission Module

The OD request module allows students to submit OD applications digitally through an online form interface.

The request form collects:

- Student information
- Event details
- Event duration
- Venue details
- Organizer information
- OD category

The request data is validated and transmitted to backend APIs for workflow processing.

Features

- Digital request submission
- Form validation
- Workflow routing
- Backend synchronization

Figure 5.4: OD Request Submission Interface

The figure displays the digital OD request form used for workflow automation and online request p

The screenshot shows the 'New OD Request' form in the ODEX system. The interface includes a dark blue sidebar with navigation options: Dashboard, My OD Requests, New OD Request (highlighted), Attendance, Calendar (with a notification badge), Profile, and Help & Support. The main content area is titled 'New OD Request' and contains the 'OD Request Submission Form'. The form is divided into four sections: 1. Student Information (Student Name: Arjun Sharma, Registration Number: IFIM2022CSI23, Department: Computer Science Engineering, Semester: 6th), 2. Event Details (Event Title / Purpose: Tech Fest 2024, Event Type: Technical Event, Event Level: College Level, Event Start Date: 20/05/2024, Event End Date: 21/05/2024, Total OD Days: 2, Venue: IFIM College Carr), 3. Organizer Details (Organizer / Company: IFIM College, Organizer Contact Person: Prof. Rahul Verma, Contact Number: 080-12345678, Organizer Email: techfest@), and 4. Additional Information (Description / Remarks: Participating in the technical event as a volunteer and presenter, Upload Supporting Document (if any)).

5.4.4 My OD Requests Module

The “My OD Requests” module allows students to monitor all submitted requests and their approval stages.

The module displays:

- Pending requests
- Approved requests
- Rejected requests
- Workflow timelines
- Approval progress

The backend dynamically updates request status during workflow execution.

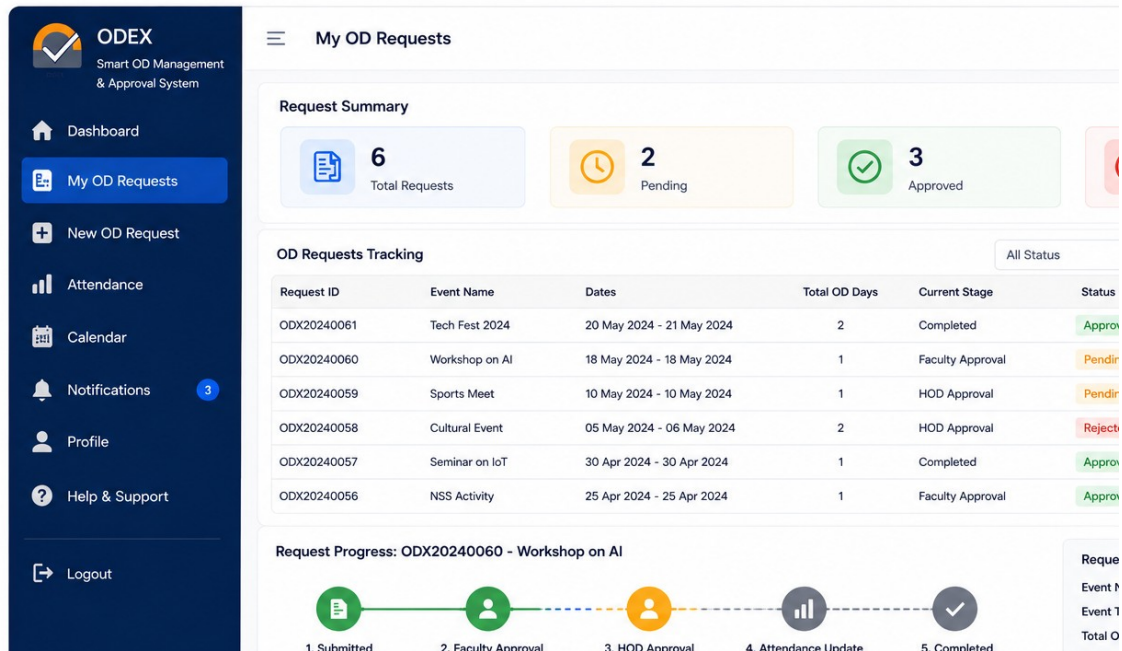
Features

- Real-time request tracking
- Approval visualization
- Workflow monitoring

- Dynamic status updates

Figure 5.5: OD Request Tracking Interface

The figure shows the dashboard used for tracking OD request approval stages and workflow pro



5.4.5 Attendance Dashboard Module

The attendance dashboard provides attendance monitoring integrated with OD approval workflows.

The module displays:

- Overall attendance percentage
- Subject-wise attendance
- Attendance analytics
- OD attendance credits
- Attendance alerts

Attendance records are updated automatically after approval completion.

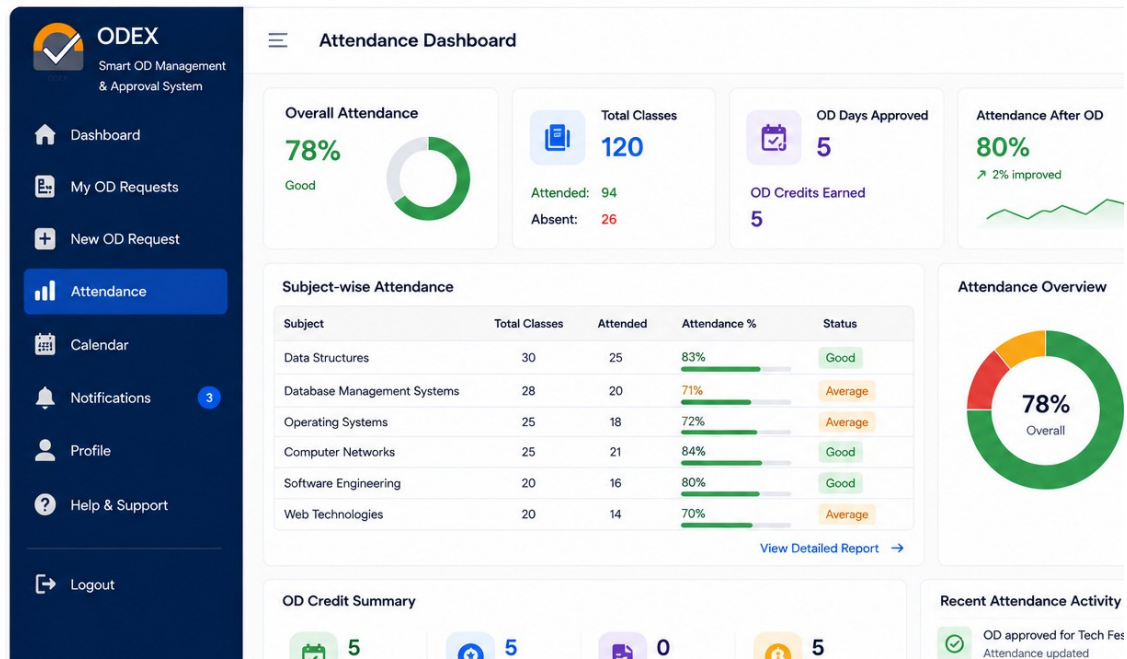
Features

- Attendance visualization
- Subject-wise analytics

- OD attendance updates
- Real-time attendance monitoring

Figure 5.6: Attendance Dashboard Interface

The attendance dashboard displays attendance statistics, OD credits, and subject-wise attendance



5.5 Backend Workflow Architecture

The backend architecture acts as the central processing engine of the ODEX platform.

The backend performs:

- Authentication processing
- API communication
- Workflow automation
- Request validation
- Attendance synchronization
- Approval routing

The Flask backend receives requests from frontend dashboards and processes workflow operations dynamically.

5.5.1 Backend Processing Layers

The backend architecture contains multiple processing layers.

Authentication Layer

Handles:

- User validation
- Role verification
- Session management

Request Handling Layer

Handles:

- API requests
- Input validation
- Workflow processing
- Error handling

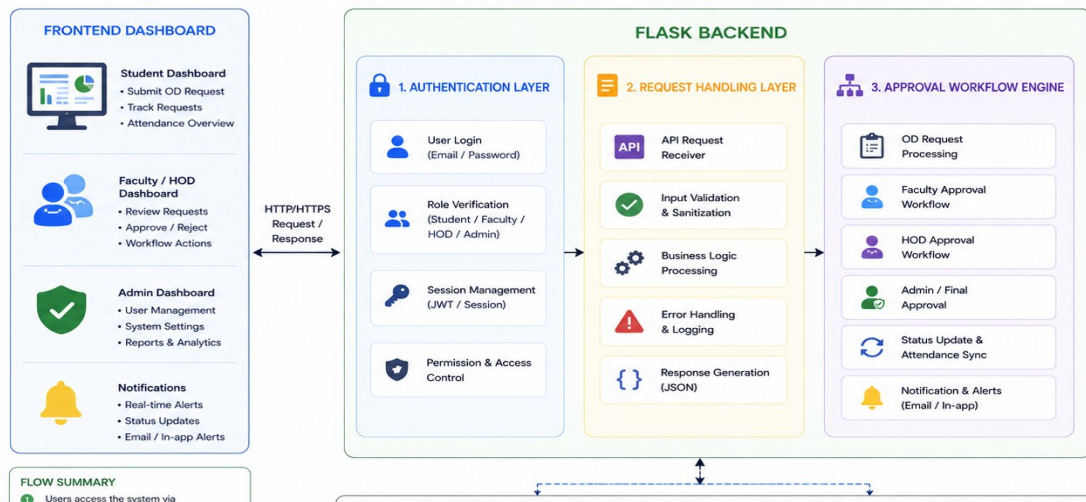
Approval Workflow Engine

Handles:

- Faculty approval
- HOD approval
- Attendance updates
- Final workflow completion

Figure 5.7: Backend Workflow Architecture

Frontend Dashboard → Flask Backend → APIs → Database



Include:

- Authentication Layer
- Request Handling Layer
- Approval Workflow Engine

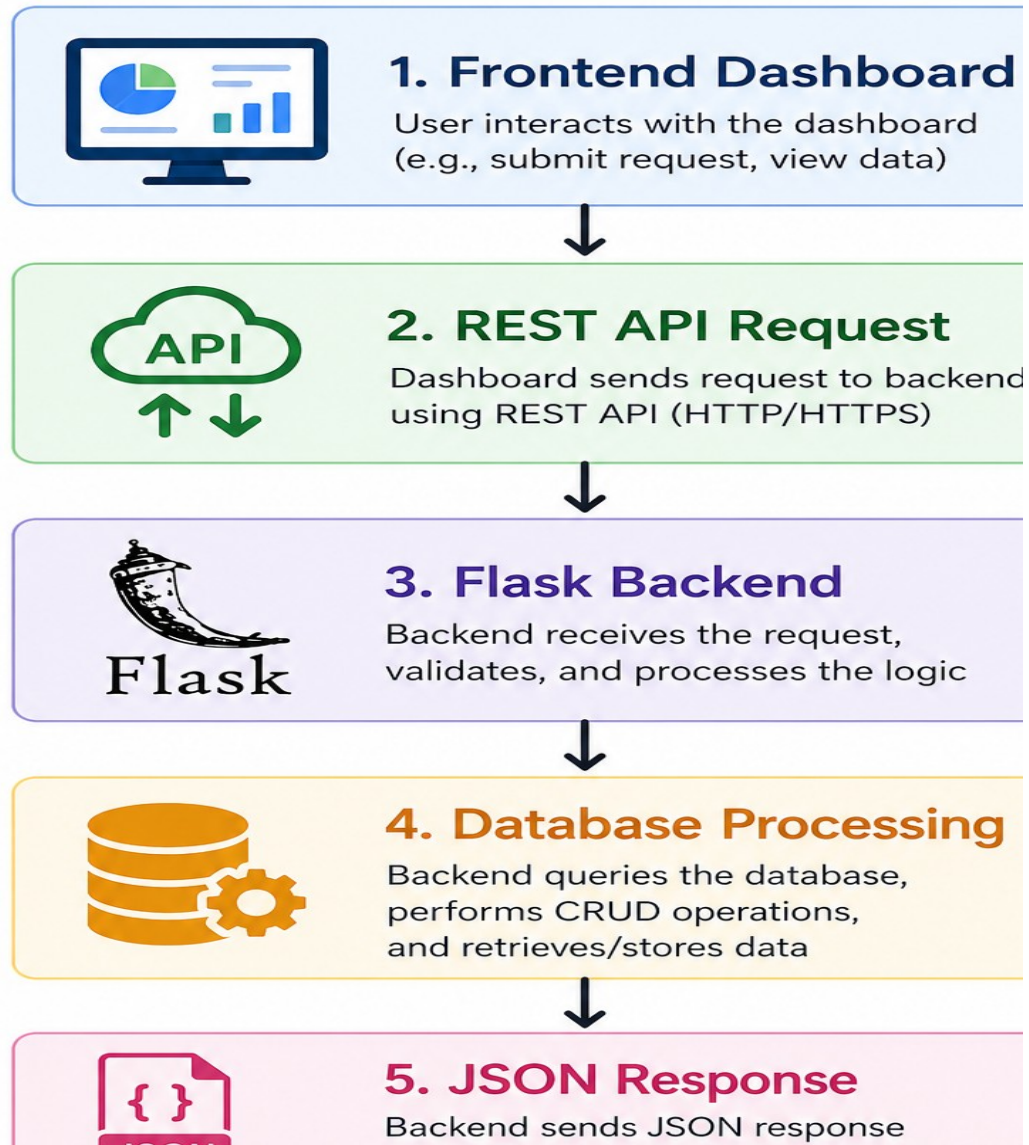
5.6 REST API Communication

REST APIs are used for communication between frontend dashboards and backend services.

The APIs support:

- Login requests
- OD request submission
- Attendance updates
- Approval processing
- Notification handling

JSON format is used for request and response communication.



5.7 Approval Workflow Engine

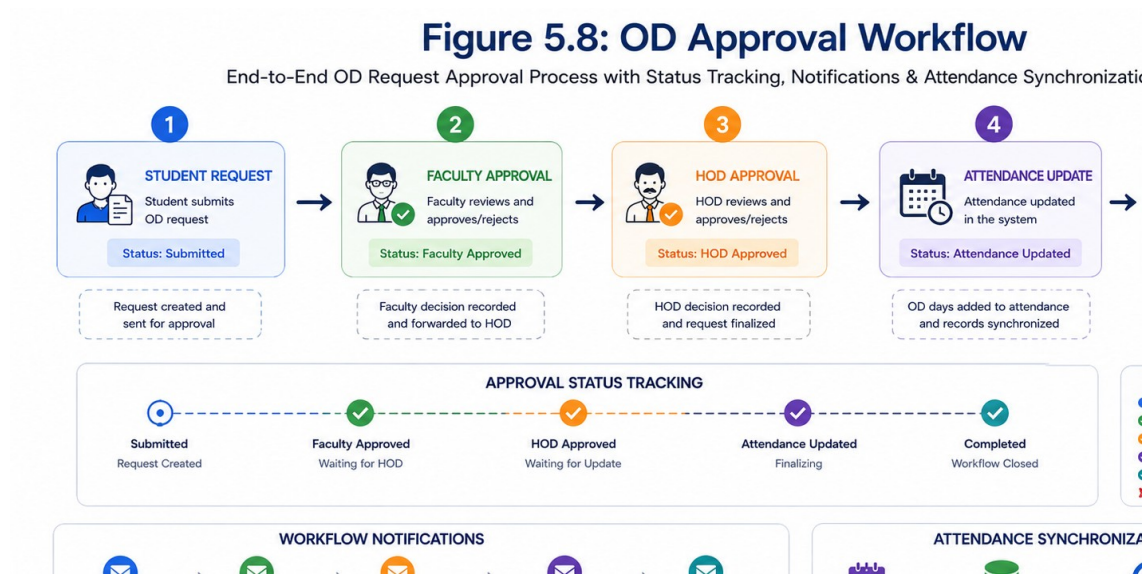
The approval workflow engine automates OD request processing between faculty members, HODs, and administrators.

The workflow sequence includes:

1. Student submits OD request
2. Faculty reviews request
3. HOD approval processing
4. Attendance update

5. Final status generation

The backend dynamically updates workflow status during each approval stage.



5.8 Database Design

The PostgreSQL database stores all workflow information centrally.

The database stores:

- User details
- Roles
- OD requests
- Attendance records
- Approval history
- Notifications

Centralized database management improves workflow synchronization and record maintenance.

5.8.1 Database Tables

Table Name	Purpose
users	Stores user information

roles	Stores role details
od_requests	Stores OD request data
approvals	Stores workflow approvals
attendance	Stores attendance records
notifications	Stores system notifications

5.9 Frontend–Backend Integration

Frontend-backend integration enables communication between React dashboards and Flask backend APIs.

The integration process includes:

- API request generation
- Backend request processing
- Database communication
- JSON response generation
- Dashboard synchronization

The integration architecture supports:

- Workflow automation
- Attendance updates
- Authentication systems
- Real-time dashboard updates

5.10 Advantages of the System

The ODEX platform provides several advantages:

- Faster workflow automation
- Reduced paperwork
- Improved transparency

- Real-time request tracking
- Automated attendance updates
- Centralized dashboard management
- Better workflow communication
- Improved authentication security

The system significantly improves institutional workflow efficiency.

5.11 Summary

This chapter explained the implementation and workflow architecture of the ODEX platform.

The project successfully integrates:

- React frontend dashboards
- Flask backend APIs
- REST communication
- Attendance synchronization
- Workflow automation
- Centralized database management

The implemented architecture improves workflow transparency, request processing efficiency, attendance management, and communication between institutional departments through centralized digital workflow automation.

CHAPTER 6: SYSTEM TESTING

6.1 Introduction

System testing is an important phase in software development used to verify whether the ODEX (Smart OD Management & Approval System) works correctly according to the project requirements.

Testing helps identify:

- Functional errors
- Workflow issues
- Authentication problems
- API communication errors
- Frontend-backend integration issues

The ODEX system was tested to ensure smooth workflow automation, attendance synchronization, dashboard communication, and approval processing.

The testing phase verified:

- Login authentication
- OD request submission
- Approval workflows
- Attendance updates
- REST API communication
- Dashboard synchronization

Proper testing improved system reliability, security, and performance.

6.2 Objectives of Testing

The major objectives of testing are:

- To verify system functionality

- To detect errors and bugs
- To ensure proper workflow processing
- To validate frontend-backend communication
- To verify attendance synchronization
- To test authentication systems
- To improve system reliability

Testing ensures that the ODEX platform functions efficiently under different workflow conditions.

6.3 Types of Testing

Different testing methods were used during the development of the ODEX platform.

6.3.1 Unit Testing

Unit testing verifies individual modules and backend functions independently.

The following modules were tested:

- Login validation
- Request submission APIs
- Attendance calculations
- Workflow processing
- Database operations

Unit testing helped identify backend logic errors during development.

6.3.2 Integration Testing

Integration testing verifies communication between frontend dashboards and backend APIs.

The testing included:

- API request handling
- JSON response validation
- Database communication

- Workflow synchronization

This testing ensured smooth frontend-backend integration.

6.3.3 System Testing

System testing verifies whether the complete ODEX platform works correctly as an integrated system.

The testing validated:

- Dashboard functionality
- Approval workflows
- Attendance synchronization
- Request tracking
- Authentication systems

System testing confirmed successful workflow automation.

6.3.4 User Interface Testing

User interface testing verifies frontend dashboard usability and navigation.

The testing included:

- Login interface testing
- Dashboard navigation
- Form validation
- Attendance dashboard rendering
- Workflow visualization

The frontend dashboards were tested using Google Chrome browser.

6.3.5 Authentication Testing

Authentication testing verifies secure user login and role-based access control.

The testing validated:

- Student login

- Faculty login
- HOD login
- Admin login
- Session validation
- Password verification

This testing improved dashboard security and authorization management.

6.3.6 Database Testing

Database testing verifies correct storage and retrieval of workflow information.

The testing included:

- User record storage
- OD request management
- Attendance updates
- Approval history tracking
- Notification storage

Database testing improved workflow synchronization and data consistency.

6.4 Test Cases

The following test cases were performed during system testing.

Table 6.1: Login Authentication Testing

Test Case ID	Test Scenario	Expected Result	Status
TC01	Valid Student Login	Dashboard Access Granted	Pass
TC02	Invalid Password	Error Message Displayed	Pass
TC03	Faculty Login	Faculty Dashboard	Pass

		Opened	
TC04	HOD Login	HOD Dashboard Opened	Pass
TC05	Admin Login	Admin Dashboard Opened	Pass

Table 6.2: OD Request Submission Testing

Test Case ID	Test Scenario	Expected Result	Status
TC06	Submit Valid OD Request	Request Stored Successfully	Pass
TC07	Empty Form Submission	Validation Error Displayed	Pass
TC08	Invalid Date Entry	Error Message Generated	Pass
TC09	File Upload Validation	File Uploaded Successfully	Pass

Table 6.3: Testing Tools Used

The following tools were used during testing:

Tool	Purpose
Google Chrome	Frontend Testing
Postman	API Testing
VS Code	Debugging
Flask Server	Backend Testing
PostgreSQL	Database Testing

6.5 Error Handling

The ODEX system includes proper error-handling mechanisms to improve workflow reliability.

The system handles:

- Invalid login credentials
- Empty form submission
- API communication errors
- Database connection issues
- Unauthorized dashboard access

Validation messages and notifications are displayed to users dynamically.

6.6 Performance Testing

Performance testing verifies system efficiency under different workflow operations.

The ODEX platform successfully handled:

- Multiple dashboard requests
- API communication
- Attendance synchronization
- Workflow processing
- Request tracking

The backend APIs responded efficiently during testing.

6.7 Security Testing

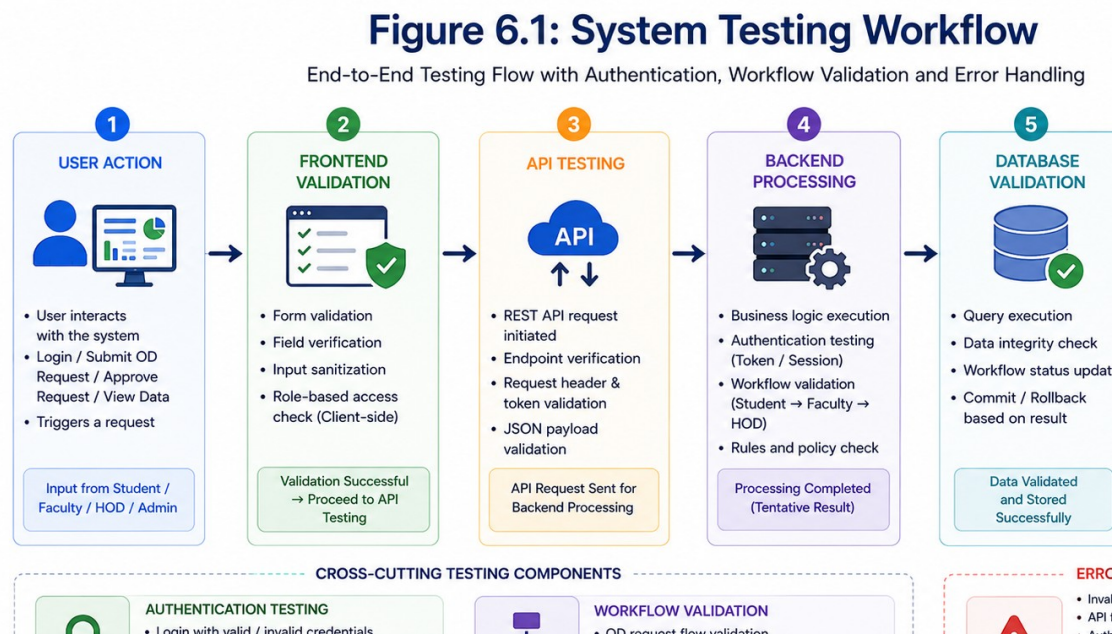
Security testing verifies authentication protection and secure workflow communication.

The testing included:

- Password validation
- Role-based authorization

- Session management
- Input validation
- API authentication

Security testing improved protection of institutional workflow data.



6.8 Summary

This chapter explained the testing methods used for verifying the ODEX platform functionality and workflow automation.

Different testing methods such as:

- Unit testing
- Integration testing
- System testing
- Authentication testing
- Database testing

were performed successfully.

The testing phase confirmed that the ODEX platform supports:

- Secure authentication
- Workflow automation
- Attendance synchronization
- Dashboard communication
- Real-time request tracking

The implemented testing procedures improved system reliability, performance, and workflow efficiency.

CONCLUSION

The ODEX (Smart OD Management & Approval System) was successfully developed to automate the traditional On-Duty approval workflow used in educational institutions. The system provides a centralized digital platform for students, faculty members, HODs, and administrators to manage OD requests efficiently through frontend dashboards and backend workflow processing.

The project successfully integrates React frontend dashboards, Flask backend APIs, REST communication, authentication systems, attendance synchronization, and centralized database management. The implemented workflow automation reduces paperwork, minimizes approval delays, improves transparency, and simplifies institutional workflow management.

The ODEX platform provides several important features such as:

- Secure role-based authentication
- Digital OD request submission
- Real-time workflow tracking
- Attendance dashboard integration
- Approval workflow automation
- REST API communication
- Centralized database management

The frontend-backend architecture successfully supports communication between dashboards, APIs, and database systems using JSON data exchange and workflow synchronization.

The implemented approval workflow engine automates request routing between faculty members, HODs, and administrators while dynamically updating attendance records and workflow status. The attendance dashboard provides subject-wise attendance monitoring and OD credit synchronization for improved academic tracking.

System testing confirmed that the platform functions efficiently under different workflow operations. Different testing methods such as unit testing, integration testing, authentication

testing, and database testing were performed successfully to improve reliability, security, and performance.

Overall, the ODEX platform successfully achieves its objective of providing a scalable, secure, and efficient OD workflow automation system capable of improving institutional communication, workflow transparency, attendance management, and approval processing through centralized digital transformation.

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